

Annotated Bibliography

Machine-Learning Forecasting of Solar Irradiance and Reliability-Constrained Off-Grid Solar Sizing Across Ghana's Climatic Zones

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The nineteen sources below ground the capstone project in five bodies of literature: (1) machine-learning and deep-learning methods for solar-irradiance forecasting; (2) statistical methodology for forecast evaluation and inference under temporal dependence (forecast-comparison tests, small-sample corrections, and the block bootstrap); (3) validation of the NASA POWER reanalysis data source for Ghana; (4) the Ghanaian energy-policy and consumption-statistics context that motivates the work and defines its research gap; and (5) off-grid photovoltaic system sizing, which underpins the project's household-application component. Annotations are analytical: each notes the source's contribution and its specific relevance to the project's research questions (RQ1–RQ5).

Ahmed, R., Sreeram, V., Mishra, Y., & Arif, M. D. (2020). A review and evaluation of the state-of-the-art in PV solar power forecasting: Techniques and optimization. *Renewable and Sustainable Energy Reviews*, 124, Article 109792. <https://doi.org/10.1016/j.rser.2020.109792>

This comprehensive review evaluates the full landscape of solar-forecasting techniques — statistical, machine-learning, deep-learning, and hybrid — across forecast horizons and input types, and catalogues the evaluation metrics (RMSE, MAE, R^2) and pre-processing strategies used in the field. The authors conclude that no single method dominates all horizons: tree ensembles and neural networks excel at short horizons while classical methods remain competitive for longer aggregation. Its systematic comparison framework directly informs this project's model-selection and evaluation design (RQ1 and RQ3), and its discussion of horizon-dependent accuracy motivates evaluating both 1-day and 7-day forecasts. As a highly cited, recent review in a leading energy journal, it provides an authoritative methodological anchor, though its focus on PV power output (rather than irradiance) means its findings are adapted rather than adopted wholesale.

Asiedu, S. T., Nyarko, F. K. A., Boahen, S., Effah, F. B., & Asaaga, B. A. (2024). Machine learning forecasting of solar PV production using single and hybrid models over different time horizons. *Heliyon*, 10(7), Article e28898. <https://doi.org/10.1016/j.heliyon.2024.e28898>

This KNUST-based study is the closest existing Ghanaian precedent for the capstone. Using operational data from a 180 kWp grid-connected PV plant at Tema, the authors compare single and hybrid machine-learning models (artificial neural network, random forest, XGBoost, and combinations) for forecasting PV production at day-ahead through month-ahead horizons, finding the ANN best for day-ahead prediction and random forest or hybrid models best at longer horizons. Its importance to this project is chiefly in defining the research gap: it demonstrates that Ghanaian ML solar forecasting exists but is confined to a single coastal industrial site and to plant power output, with no multi-zone comparison, no seasonal skill stratification, and no household application. The capstone extends exactly these dimensions

(RQ1, RQ4, RQ5). Methodologically, its horizon-dependent model rankings inform the project's expectation that different model families may win at one-day versus seven-day horizons (RQ3).

Asumadu-Sarkodie, S., & Owusu, P. A. (2016). The potential and economic viability of solar photovoltaic power in Ghana. *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects*, 38(5), 709–716. <https://doi.org/10.1080/15567036.2015.1122682>

This study assesses the technical and economic feasibility of solar photovoltaic deployment in Ghana, documenting the country's solar-resource endowment (daily irradiation of roughly 4–6 kWh/m²) and arguing that solar PV is a viable complement to hydro-dominated generation. The authors highlight the north–south resource gradient, with the strongest and most stable resource in northern Ghana. For this project, the paper establishes the domain motivation: it demonstrates commercial interest in Ghanaian solar and justifies the choice of study sites spanning the coastal, forest, and savannah zones (RQ1 and RQ4). Its economic framing also supports the recommendations section, connecting forecast accuracy to investment risk. The study predates modern ML forecasting and uses static resource assessment, which is precisely the gap this capstone addresses — moving from average resource maps to dynamic, model-based prediction.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>

This is the foundational paper for one of the project's two core tree-ensemble models. Breiman introduces random forests — ensembles of decorrelated decision trees built on bootstrap samples with random feature selection at each split — and proves that generalization error converges as trees are added, making the method resistant to overfitting. Two properties matter directly for this capstone. First, random forests handle nonlinear interactions among the engineered predictors (lagged GHI, rolling means, calendar harmonics, meteorological drivers) without distributional assumptions, supporting RQ1 and RQ3. Second, the paper's variable-importance machinery is the ancestor of the permutation-importance analysis used in RQ2. A known limitation is directly relevant to forecasting: tree ensembles cannot extrapolate beyond the range of their training targets, which motivates reporting seasonal error stratification (RQ4) to check behavior under extreme low-irradiance conditions. Citing the primary source anchors the method choice academically rather than resting on software defaults.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM. <https://doi.org/10.1145/2939672.2939785>

This paper introduces XGBoost, the gradient-boosted tree implementation that has become a de facto standard for structured-data prediction, describing its regularized objective, sparsity-

aware split finding, and computational optimizations. It is included as the primary methodological citation for one of the project's core models. XGBoost is well suited to this capstone because engineered tabular features (lags, rolling means, cyclical calendar encodings, meteorological drivers) are exactly the input type where boosted trees are most competitive, and its native feature-importance outputs feed the driver analysis in RQ2. The paper's regularization discussion also informs hyperparameter tuning choices (depth, learning rate, subsampling) in the project pipeline. As a peer-reviewed systems paper with tens of thousands of citations, it is a credible foundation, though model-specific performance claims will be re-validated on the Ghana data rather than assumed.

Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. <https://doi.org/10.1080/07350015.1995.10524599>

This paper supplies the formal statistical machinery for RQ3, the project's central inferential question. Diebold and Mariano show how to test whether two forecasts differ in accuracy by working directly with the loss-differential series — the day-by-day difference in (squared) forecast errors — and constructing a test statistic whose long-run variance accounts for the autocorrelation that forecast errors inevitably exhibit. This is exactly the property the peer-reviewed design requires: daily GHI forecast errors are serially dependent, so naive paired t-tests overstate significance. The test is loss-function-flexible (squared or absolute error) and requires no assumption that models are correctly specified. In this project it is applied one-sided (ML versus classical benchmarks), with the Harvey–Leybourne–Newbold small-sample correction and Holm adjustment across the family of model-site-horizon comparisons. Its known caveat — distorted size when comparing nested models — is acknowledged, which is one reason the moving-block-bootstrap interval serves as a complementary check.

Energy Commission of Ghana. (2019). *Ghana renewable energy master plan*.

<https://www.energycom.gov.gh/files/Renewable-Energy-Masterplan-February-2019.pdf>

The Renewable Energy Master Plan (REMP) is Ghana's official policy roadmap for renewable deployment through 2030, targeting an increase in renewable capacity from a 42.5 MW baseline (2015) to about 1,364 MW, including roughly 500 MW of utility-scale and 200 MW of distributed solar, with an estimated investment need of USD 5.6 billion. As a primary government source, it defines the stakeholder context for this capstone: the utilities, regulators, and developers who must integrate the planned solar capacity are the end-users of irradiance forecasts. The REMP quantifies the scale of the planning problem and anchors the project's recommendations (Section 10 of the synopsis) in stated national policy rather than hypothetical demand. Being a policy document, it contains no forecasting methodology; it is used strictly for motivation, targets, and stakeholder identification rather than technical guidance.

Energy Commission of Ghana. (2025). *2025 national energy statistical bulletin*.
<https://www.energycom.gov.gh/planning/energy-statistics>

The Energy Commission's annual statistical bulletin is the official record of Ghana's energy production, consumption, and customer numbers, and it anchors the demand side of RQ5. The bulletin's figures — residential consumption on the order of 8,569 GWh in 2024 across roughly 5.28 million residential customers — imply an average of about 135 kWh per customer per month, which grounds the study's 200 kWh/month benchmark as a moderate-to-high consumption scenario rather than a national median. Two interpretive cautions are built into the study's use of this source: 'customers' are metered connections rather than households (compound houses commonly share one meter, so per-household use is lower), and the consumption distribution is right-skewed given the substantial lifeline-band population. The specific figures will be re-verified against the bulletin's primary tables during the study, and the three consumption tiers (100/200/300 kWh per month) make the sizing results robust to categorization differences in this statistic.

Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. [https://doi.org/10.1016/S0169-2070\(96\)00719-4](https://doi.org/10.1016/S0169-2070(96)00719-4)

Harvey, Leybourne, and Newbold show that the original Diebold–Mariano test is oversized in moderate samples — it rejects the null of equal accuracy too often — and derive a modified statistic with a small-sample correction factor, compared against the Student-t rather than normal distribution. This correction matters concretely for the capstone: per-site forecast comparisons rest on roughly 730 test days, and even the pooled comparison (about 3,650 paired days) involves autocorrelated loss differentials at the 7-day horizon, where forecast errors overlap mechanically. Applying the HLN correction therefore protects RQ3's conclusions from spurious significance — a direct response to peer-review feedback on the study's statistical design. The paper also motivates the study's broader posture of treating time-series inference conservatively, pairing corrected parametric tests with moving-block-bootstrap intervals. It is short, technical, and standard; its role in the bibliography is methodological insurance rather than domain content.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>

The original LSTM paper introduces the gated recurrent architecture — input, output, and (in later variants) forget gates around a constant-error memory cell — that solves the vanishing-gradient problem preventing ordinary recurrent networks from learning long temporal dependencies. It is cited as the primary source for the project's deep-learning forecaster, which consumes 30-day windows of GHI and meteorological history (RQ1, RQ3). The architecture's

strength for this application is its ability to carry information across multi-day weather episodes, such as Harmattan dust sequences, that fixed-lag tabular models summarize only through hand-built features. The paper also frames the project's realism about deep learning: LSTMs are data-hungry and sensitive to architecture and initialization, and with only about 5,800 training days per site the study does not presume LSTM superiority — hence the pooled multi-site variant, repeated seeded runs, early stopping, and the expectation that tree ensembles may prove competitive, as tropical benchmarking studies suggest.

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>

This widely used open-access textbook provides the statistical foundations for the project's classical baselines and evaluation design: seasonal-naïve methods, SARIMA modelling, time-series decomposition, and — critically — correct out-of-sample evaluation using temporal train/test splits and time-series cross-validation. Its treatment of forecast-accuracy measures and the pitfalls of information leakage directly shapes the project's temporal-split design (train 2005–2020, validate 2021–2022, test 2023–2024) and the choice of RMSE/MAE as primary metrics (RQ1, RQ3). The book's authority (standard reference in forecasting courses worldwide) and practical orientation make it an appropriate methodological source. Because it is a general text rather than a solar-specific study, it is complemented by the domain-specific reviews (Voyant et al., 2017; Ahmed et al., 2020) for solar-particular modelling choices.

Khatib, T., Mohamed, A., & Sopian, K. (2013). A review of photovoltaic systems size optimization techniques. *Renewable and Sustainable Energy Reviews*, 22, 454–465. <https://doi.org/10.1016/j.rser.2013.02.023>

This review is the methodological foundation for the project's off-grid application (RQ5). It surveys standalone photovoltaic sizing techniques — intuitive rules of thumb, numerical (simulation-based) methods, analytical formulations, and hybrid approaches — and frames system reliability through the loss-of-load-probability (LOLP) concept: the fraction of time a PV-battery system fails to serve the load. The authors show that numerical simulation over long historical irradiance records is the most accurate sizing approach, and that sizing outcomes are highly location-dependent, which is precisely the premise of the capstone's zone-specific sizing tables for Ghanaian households. The project implements the review's recommended approach: a daily energy-balance simulation driven by twenty years of NASA POWER data per climatic zone, searching array/battery combinations that meet 1% and 5% LOLP targets for a 200 kWh/month household. The review predates modern ML, which the capstone layers on top through forecast-informed load-shifting warnings.

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241. <https://doi.org/10.1214/aos/1176347265>

Künsch's moving-block bootstrap is the inferential backbone of the study's uncertainty quantification. The ordinary bootstrap assumes independent observations; resampling autocorrelated daily data point-by-point destroys the very dependence structure that makes solar time series (and their forecast errors) behave as they do, producing intervals that are too narrow. Künsch's solution — resampling contiguous blocks of observations so that within-block dependence is preserved — yields asymptotically valid inference for stationary dependent data. In this project the method underpins the confidence intervals for RMSE, MAE, R^2 , and skill on the test window (RQ1), the dependence-robust check on Diebold–Mariano conclusions (RQ3), and, critically, the interval estimates for LOLP in the sizing backtest (RQ5), where loss-of-load days cluster in runs and an independent-Bernoulli calculation would overstate precision — the exact issue raised in peer review of the sample-size section. Year-block resampling is used as a coarser complement for seasonal statistics.

Lauret, P., Voyant, C., Soubdhan, T., David, M., & Poggi, P. (2015). A benchmarking of machine learning techniques for solar radiation forecasting in an insular context. *Solar Energy*, 112, 446–457. <https://doi.org/10.1016/j.solener.2014.12.014>

This benchmarking study compares neural networks, Gaussian processes, and support-vector regression against classical autoregressive baselines for hourly solar forecasting on tropical islands (Corsica, Guadeloupe, Réunion), finding that ML methods deliver their largest gains under unstable, cloudy sky conditions while simple baselines remain competitive under clear skies. Its central insight — that the value of ML depends on sky-condition regime — maps directly onto this project's RQ4, which asks whether forecast accuracy in Ghana differs between wet, dry, and Harmattan seasons. The tropical setting makes its findings more transferable to Ghana than temperate-latitude studies. The study also demonstrates rigorous paired benchmarking against naive baselines, the same design used for RQ3. Its hourly resolution differs from this project's daily resolution, so effect sizes are expected to differ, but the comparative methodology carries over cleanly.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30* (pp. 4765–4774). Curran Associates.

This paper introduces SHAP (SHapley Additive exPlanations), the model-interpretation framework used to answer RQ2. Lundberg and Lee unify several earlier attribution methods by showing they are all approximations of Shapley values from cooperative game theory — the unique additive attribution satisfying local accuracy, missingness, and consistency — and provide efficient estimators for tree ensembles. In this project, SHAP values rank the contribution of each meteorological and calendar predictor to the GHI forecasts,

complementing correlation analysis with a view of what the fitted models actually use, and enabling zone-wise comparison of predictor importance. The study adopts the interpretive discipline the peer review demanded and the paper itself implies: SHAP explains predictions conditional on a specific model and feature set, so its outputs are reported as model-specific predictive contributions — evidence of association and predictive value, not causal effects. This caveat is stated explicitly in the synopsis's RQ2 hypotheses.

NASA Langley Research Center. (2024). *NASA prediction of worldwide energy resources (POWER)* [Data set]. <https://power.larc.nasa.gov/>

NASA POWER is the project's primary data source: a free, openly accessible archive of daily satellite-derived solar and meteorological parameters (including all-sky surface shortwave irradiance, temperature, humidity, cloud amount, precipitation, and wind speed) available globally via a documented REST API with no registration requirement. The Renewable Energy community collection is specifically curated for solar-energy applications. For this capstone, POWER supplies twenty years (2005–2024) of daily observations for all five Ghanaian study sites, enabling the sample sizes computed in the synopsis and guaranteeing full reproducibility, since any reader can re-download identical data from the cited endpoints. As a reanalysis/satellite product, it carries known biases relative to ground pyranometers — the reason the Quansah et al. (2022) validation study is included — and its $\sim 0.5^\circ$ grid resolution limits fine-scale siting conclusions, a limitation acknowledged in the synopsis.

Qing, X., & Niu, Y. (2018). Hourly day-ahead solar irradiance prediction using weather forecasts by LSTM. *Energy*, 148, 461–468. <https://doi.org/10.1016/j.energy.2018.01.177>

This influential paper applies long short-term memory (LSTM) networks to day-ahead solar-irradiance prediction using weather variables as inputs, reporting an 18–34% RMSE improvement over multilayer-perceptron and persistence baselines on data from Santiago de Cuba — again a tropical setting. It demonstrates that LSTM's ability to learn temporal dependencies yields concrete accuracy gains for exactly the day-ahead horizon this project targets, and its input design (meteorological forecasts plus calendar features) closely parallels the project's feature set. The paper is the main methodological precedent for the project's LSTM component (RQ1, RQ3) and its reported effect sizes inform the expectation that ML can beat classical baselines by the project's 15% RMSE success threshold. Limitations include a single study site and hourly resolution; this capstone extends the design to multiple climatic zones at daily resolution with formal significance testing.

Quansah, A. D., Dogbey, F., Asilevi, P. J., Boakye, P., Darkwah, L., Oduro-Kwarteng, S., Sokama-Neuyam, Y. A., & Mensah, P. (2022). Assessment of solar radiation resource from the NASA-POWER

reanalysis products for tropical climates in Ghana towards clean energy application. *Scientific Reports*, 12, Article 10684. <https://doi.org/10.1038/s41598-022-14126-9>

This is the project's most critical data-validation source. The authors compare NASA POWER solar-radiation estimates against sunshine-duration-based ground estimates at 22 synoptic stations across Ghana, finding correlations of $r = 0.60\text{--}0.94$, with the strongest agreement over the northern savannah (especially March–May) and the weakest over the southern forest zone during June–August. The study concludes that POWER data are a credible long-term solar-resource reference for Ghana. For this capstone the paper does double duty: it justifies the choice of NASA POWER as the primary dataset (Section 5 of the synopsis), and its spatial pattern of agreement directly informs the interpretation of zone-level results (RQ1, RQ4) — model errors in the forest zone must be read against higher underlying data uncertainty there. Published in a peer-reviewed Nature Portfolio journal by Ghana-based researchers, it is both credible and directly on-point.

Voyant, C., Notton, G., Kalogirou, S., Nivet, M.-L., Paoli, C., Motte, F., & Foulloy, A. (2017). Machine learning methods for solar radiation forecasting: A review. *Renewable Energy*, 105, 569–582. <https://doi.org/10.1016/j.renene.2016.12.095>

This heavily cited review synthesizes the machine-learning literature on solar-radiation forecasting, covering artificial neural networks, support-vector machines, tree ensembles, and hybrid approaches, and comparing their reported performance across horizons and climates. Two of its conclusions shape this project's design: first, ensemble and hybrid methods tend to outperform single learners, supporting the inclusion of random forest and XGBoost alongside the LSTM (RQ1, RQ3); second, the review stresses that accuracy comparisons are only meaningful against common baselines on common data — the rationale for this project's shared temporal hold-out and paired statistical testing. The review also catalogues standard input features (lagged irradiance, meteorological variables, calendar encodings) that the project's feature-engineering step implements. Published before the recent deep-learning wave, it slightly underrepresents modern architectures, which the Qing and Niu (2018) source complements.